

Nicole Leow Ke Xin

0458686003

nicole.leow@hotmail.com

Summary

Driven Mechatronics Engineering and Computer Science student at UNSW, achieving a High Distinction WAM. Experienced in software development, project management, critical thinking and collaborative teamwork through roles at Telstra and UNSW. A proactive problem-solver with strong communication and adaptability skills, always eager to learn and contribute to cutting-edge projects. Passionate about technology and committed to continuous growth in a dynamic and innovative environment.

Education

UNSW Sydney (2021 - present)

- Bachelor of Mechatronics Engineering (Honours) and Bachelor of Science (Computer Science)
- High Distinction WAM
- International Scientia Coursework Scholarship (full tuition fee remission)
- Faculty of Engineering Dean's List 2022, 2023 and 2024

Sunway College (2020 - 2021)

- 4A* (High Distinction) in the Cambridge A Levels Physics, Chemistry, Biology and Maths

Technical Skills

- Programming: C, C++, Java, JavaScript, Python
- Web Development: React, Express.js, HTML, CSS, TypeScript
- Databases: PostgreSQL, MongoDB
- Networks and Applications
- Engineering Tools: Fusion 360, SolidWorks, MATLAB
- Manufacturing: 3D Printing, Laser Cutting

Experiences

UNSW Casual Academic (2024 - Current)

- Tutor for four undergraduate courses in programming (OOP, Java, C++) and engineering.
- Managed and led classes of engineering students, guiding them through core concepts and practical applications.
- Fostered a supportive and interactive learning environment.

Telstra Undergraduate Software Engineer (2024 - Current)

- Analysed and corrected data errors within the Digital Video Network, improving quality assurance and data integrity.
- Collaborated with teams in an agile environment to update data processing and error detection SQL scripts.
- Designed a network inventory database and created API endpoints for various microservices using JavaScript and MongoDB.

Jane Street Insight Program Participant (2024)

- Engaged in intensive learning sessions covering probability, market structure, and arbitrage, including interactive group activities and mock trading simulations.

UNSW Mechatronics Society Executive Member (2023)

- Ensured the smooth operation of the society by overseeing events, workshops, and projects run by the society.
- Worked collaboratively with other executive members to develop and execute strategies to promote the society and attract new members.
- Coordinated with external stakeholders and professionals to establish partnerships and secure funding.

Connected Health Software Engineer (2023)

- Developed machine learning models to identify cancerous oral lesions for oral cancer removal using Python and MATLAB.

Projects

Bookings.com - Web-based Room Booking System (2024)

- Developed a full-stack room booking system using React, Express.js, TypeScript, Prisma, and PostgreSQL.
- Communicated effectively with the client to gather and validate requirements, ensuring alignment with project goals.

Presto - Web-based Presentation App (2024)

- Worked on frontend development of a web-based presentation app in ReactJS, ensuring high standards of UI/UX and accessibility while interfacing with RESTful API developed by team members.

Treats - Backend JavaScript Server (2022)

- Developed the backend server of a communication platform that allows students to communicate, and collaborate virtually
- Demonstrated effective use of static testing, dynamic testing, and user testing to validate software systems.
- Demonstrated an understanding of version control, continuous integration, and deployment tooling to sustainably integrate code from multiple parties.

Soccer Droid (2022)

- Designed and 3D printed custom robot components using CAD software.
- Integrated sensors for motion tracking and obstacle detection.
- Programmed and tested the robot using Arduino IDE for autonomous operation.